

ZACK ED DS

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Education

University of Illinois Urbana-Champaign, Grainger College of Engineering
BS/MCS 5-Year Program in Computer Science – AI & Data Focus

Master's expected May 2027

GPA: 3.96/4.00

Coursework: Deep Learning, Artificial Intelligence, Applied ML, Text Information Systems, Database Systems, Algorithms

Honors: Tau Beta Pi Engineering Honors (2x Outstanding Member), NVIDIA John Nickolls Scholar, Engineering Visionary Scholar, Dean's List

Experience

Apple Inc., Software Engineering Intern, Cupertino, CA

Peak Power Management – Battery Software

In Progress – Summer 2026

- Internship in progress.

Books & Notes – iOS Quality Engineering

May 2025 – Aug 2025

- Led pre-submission testing for the Books/Notes iOS 26 redesign, including seed and Root build sign-offs.
- Filed 28 critical P1 bugs; in the first two months ranked second of eight FTE engineers.
- Prototyped an agentic workflow via Radar, GitHub, and Device MCPs to autonomously verify fixes on-device.
- Created documentation covering 30+ Book Store elements, streamlining bug reporting and testing.

University of Illinois Urbana-Champaign, Research Assistant

LLM Agents – Prof. Lingming Zhang

Jan 2026 – Present

- Open-sourced Spec-Bench extension for agent swarms (Llama-70B + 1B); wrote canonical C++ baseline, agent hit 5.66× speedup. Stratified prompt subset cut eval wall time 3h→3min with 0.7% bias.
- Designed three context management configs; persistent sessions 3–4× cheaper than resetting loops on long runs.
- Built algorithmic analysis pipeline auditing 739 agent programs via three plot families (score progression, parent-selection, code-diversity); revealed extreme greediness in 9/10 problems and surfaced data-quality issues for paper.

AI & Computer Vision – Prof. Saurabh Gupta

Jun 2024 – Aug 2024

- Synthesized human hand-object interaction datasets in computer vision and robotics.
- Tuned HaMeR transformer model for 3D hand reconstruction on wrist-mounted camera; ablations dropped the RCNN framing stage and threshold tuning improved detection.
- Presented findings at the UIUC Research Symposium; research contributed to a paper accepted to CVPR 2025.

U.S. Department of Energy, Software Engineering Intern, Argonne National Laboratory

May 2023 – Aug 2023

- Built interactive Svelte front-end for electric vehicle charging metrics in the AFLEET carbon emissions database.
- Analyzed SQL database and wrote queries to construct electrical-rate tables.
- Identified and reported a critical flaw to the United States Utility Rate Database team.

LanzaJet Inc., Software Engineering & Finance Intern

Jun 2022 – Sep 2024

- Built Python data pipeline for commodity-market statistics using SOAP/REST APIs and web scraping with Pandas.
- Led cross-functional team to build refinery inventory system in MS Power Apps with Dataverse and AVEVA PI Vision.

Project Highlights

Terminal Agent – Fine-Tuning

- QLoRA fine-tuned 2B model via MLX on Apple Silicon; 59%→91%
- Docker eval framework: 67 test cases, 201 trials, automated scoring
- Found macOS GPU bug; filed to MLX, got undocumented fix

NoRules.fun – LLM PvP Game

- LLM referee; unlimited actions
- DeepSeek + FLUX image gen
- Built 49-test eval suite; A/B tested system prompts (39%→80%)
- Streaming pipeline; 1.5s TTFB
- ~\$0.035/game; prompt-tuned

McKinley Health Chatbot – RAG

- Flask backend with FAISS retrieval & HuggingFace hosted Llama LLM
- Scraped 700+ health center pages
- Embedded text with transformer model

Technical Skills

Languages: Python, Swift, C++, Java, JavaScript, SQL

AI/ML: PyTorch, TensorFlow, MLX, LoRA, LLM Fine-Tuning, FAISS, HuggingFace, RAG, Prompt Engineering, Agentic Systems

Tools & Frameworks: Flask, Svelte, SwiftUI, Docker, Git, Pandas, NumPy, Matplotlib, Beautiful Soup, ROS 2, Tableau

Platforms: AWS Bedrock, Google Cloud, Snowflake, OpenClaw, macOS/iOS, Linux